1 After first making a drawing to figure out the relationship of the parts and the exact sizes required to accommodate the refill, cut the Ebonite rods to length. Begin by turning a spigot on a piece of 20mm black Ebonite to form the external threads on the nib unit. There are many ways the material could be mounted on the lathe; I chose to use engineers jaws in a scroll chuck. Pen blank jaws or pin jaws would also work well or even a collet chuck

2 Using the methods outlined in my previous articles, cut the threads on the spigot using an M10 x 1.25 die. This is quite a large size and a coarse thread, but the size down is usually M8, which does not leave sufficient clearance to accommodate the 6.5mm diameter bore required for the body of the refill. The size makes for a chunky pen and the coarse thread is concealed when the pen is in use

3 Drill out for the refill, first drilling right through at 3mm diameter, then to 4.8mm and 6.5mm to the precise depths required to exactly match the shape of the rollerball refill. This will ensure a good fit of the refill in the writing tip of the pen. If your tailstock quill is calibrated you can use this to measure the depth, otherwise mark the drill with tape

4 Here is the completed thread and bore which needs a tidy up. The threads nearest to the body of the unit have been trimmed away to provide clearance so the components fit tightly together when assembled. Take care not to overcut and remove the threaded spigot or weaken it so it snaps off in use

5 Make a mandrel from scrap acrylic or Ebonite by drilling and tapping an M10 x 1.25 thread in the end. The nib section can then be mounted on this, supported by a 60° revolving centre in the tailstock and turned to shape

6 Once you are happy with the shape, sanding and polishing can also be completed with the work mounted on the mandrel. Any woodworking abrasive will be suitable for use. Work through the grits and do not use too much speed or pressure

7 With the nib unit completed, begin work on the body of the pen. With a suitably sized Ebonite rod mounted in the chuck, cut a spigot for the external thread (cap to body). For this, I used a ¼in BSP thread, often supplied in metric tap and die sets as it is a commonly used size in plumbing and compressed air fittings. It fits neatly over the 11mm diameter nib section

8 Next, drill the barrel to length for the refill (this length is critical if the pen is to write properly) and then to size for the internal thread. As there is not much material in the threaded section, it is a good idea to provide support while drilling and cutting the threads

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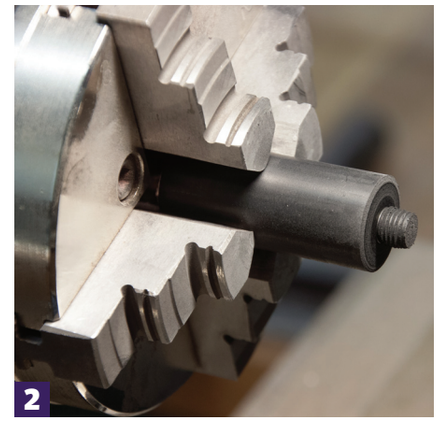
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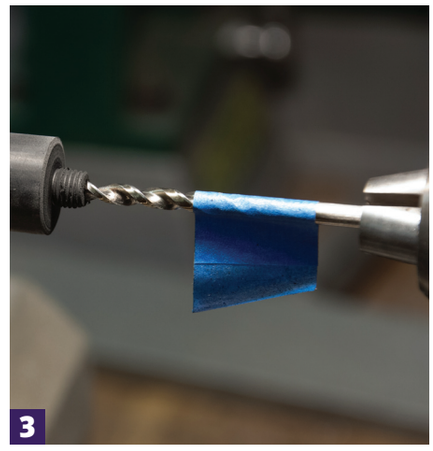
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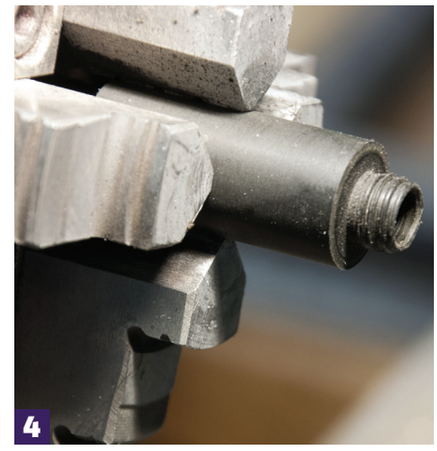
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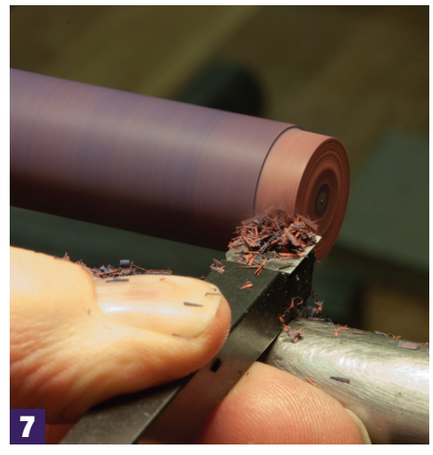
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